Project 01: Quick Lipstick Lookup (O(1))

1. BeautyLab01.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>
<BuiltInComInteropSupport>true</BuiltInComInteropS
upport>
<ApplicationManifest>app.manifest</ApplicationManif
est>
<AvaloniaUseCompiledBindingsByDefault>false</Ava
loniaUseCompiledBindingsByDefault>
  </PropertyGroup>
  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>

  <ItemGroup>
    <AvaloniaResource Include="Images/**" />
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="BeautyLab01.MainWindow"
        Title="BeautyLab ♡ Quick Lipstick Lookup"
        Width="950" Height="720"
        Background="#FFF6C9"
        WindowStartupLocation="CenterScreen">

    <ScrollViewer>
        <StackPanel Margin="28" Spacing="20">

            <Border Background="#FFD9E8"
                    CornerRadius="24" Padding="22">
                <StackPanel Spacing="8">
                    <TextBlock Text="♡ B E A U T Y L A B ♡"
                               FontSize="16" FontWeight="Bold"
                               Foreground="#78465C"
                               HorizontalAlignment="Center"/>
                    <TextBlock Text="Quick Lipstick Lookup"
                               FontSize="30" FontWeight="Bold"
                               Foreground="#78465C"
                               HorizontalAlignment="Center"/>
                    <TextBlock Text="Find your favorite
beauty product! ✨"
                               FontSize="15"
                               Foreground="#78465C"
                               HorizontalAlignment="Center"/>
                </StackPanel>
            </Border>

            <TextBlock Text="♡ CHOOSE YOUR BEAUTY
PICK"
                       FontSize="20" FontWeight="Bold"
                       Foreground="#78465C"/>

            <Grid ColumnDefinitions="*,*,*"
                  RowDefinitions="Auto,Auto"
                  ColumnSpacing="14" RowSpacing="14">

                <Border Grid.Row="0" Grid.Column="0"
                        Background="#FFFDF5"
                        CornerRadius="18" Padding="12">
                    <StackPanel Spacing="8">
                        <Image Source="/Images/lipstick.jpeg"
                               Height="140" Stretch="Uniform"/>
                        <TextBlock Text="Lipstick"
                                   FontSize="17"
FontWeight="Bold"
                                   Foreground="#78465C"
                                   HorizontalAlignment="Center"/>
                        <Button Content="Select ♡"
                                Background="#F4A7C5"
                                Foreground="#60384B"
                                Click="Lipstick_Click"/>
                    </StackPanel>
                </Border>

                <Border Grid.Row="0" Grid.Column="1"
                        Background="#FFFDF5"
                        CornerRadius="18" Padding="12">
                    <StackPanel Spacing="8">
                        <Image
Source="/Images/foundation.jpeg"
                               Height="140" Stretch="Uniform"/>
                        <TextBlock Text="Foundation"
                                   FontSize="17"
FontWeight="Bold"
                                   Foreground="#78465C"
                                   HorizontalAlignment="Center"/>
                        <Button Content="Select ♡"
                                Background="#F4A7C5"
                                Foreground="#60384B"
                                Click="Foundation_Click"/>
                    </StackPanel>
                </Border>

                <Border Grid.Row="0" Grid.Column="2"
                        Background="#FFFDF5"
                        CornerRadius="18" Padding="12">
                    <StackPanel Spacing="8">
                        <Image Source="/Images/blush.jpeg"
                               Height="140" Stretch="Uniform"/>
                        <TextBlock Text="Blush"
                                   FontSize="17"
FontWeight="Bold"
                                   Foreground="#78465C"
                                   HorizontalAlignment="Center"/>
                        <Button Content="Select ♡"
                                Background="#F4A7C5"
```

```
                    Foreground="#60384B"
                    Click="Blush_Click"/>
            </StackPanel>
        </Border>

        <Border Grid.Row="1" Grid.Column="0"
                Background="#FFFDF5"
                CornerRadius="18" Padding="12">
            <StackPanel Spacing="8">
                <Image
Source="/Images/mascara.jpeg"
                       Height="140" Stretch="Uniform"/>
                <TextBlock Text="Mascara"
                           FontSize="17"
FontWeight="Bold"
                           Foreground="#78465C"
                           HorizontalAlignment="Center"/>
                <Button Content="Select ♡"
                        Background="#F4A7C5"
                        Foreground="#60384B"
                        Click="Mascara_Click"/>
            </StackPanel>
        </Border>

        <Border Grid.Row="1" Grid.Column="1"
                Background="#FFFDF5"
                CornerRadius="18" Padding="12">
            <StackPanel Spacing="8">
                <Image
Source="/Images/lipgloss.jpeg"
                       Height="140" Stretch="Uniform"/>
                <TextBlock Text="Lip Gloss"
                           FontSize="17"
FontWeight="Bold"
                           Foreground="#78465C"
                           HorizontalAlignment="Center"/>
                <Button Content="Select ♡"
                        Background="#F4A7C5"
                        Foreground="#60384B"
                        Click="LipGloss_Click"/>
            </StackPanel>
        </Border>

        <Border Grid.Row="1" Grid.Column="2"
                Background="#FFEAB0"
                CornerRadius="18" Padding="18">
            <StackPanel Spacing="10"
                        VerticalAlignment="Center">
                <TextBlock Text="♡"
                           FontSize="38"
                           HorizontalAlignment="Center"/>
                <TextBlock Text="QUICK LOOKUP"
                           FontSize="17"
FontWeight="Bold"
                           Foreground="#78465C"
                           HorizontalAlignment="Center"/>
                <TextBlock Text="Choose a product to
display its details."
                           TextWrapping="Wrap"
                           TextAlignment="Center"
                           Foreground="#78465C"/>
            </StackPanel>
        </Border>
    </Grid>

    <Border Background="#FFFDF5"
            CornerRadius="20" Padding="20">
        <StackPanel Spacing="12">
            <TextBlock Text="♡ Selected Product"
                       FontSize="21" FontWeight="Bold"
                       Foreground="#78465C"/>
            <TextBlock x:Name="ResultText"
                       Text="Select a product to see its
details!"
                       FontSize="18"
                       Foreground="#78465C"
                       TextWrapping="Wrap"/>
            <TextBlock Text="BIG-O NOTATION:
O(1) — CONSTANT TIME"
                       FontSize="14" FontWeight="Bold"
                       Foreground="#B75E85"/>
            <TextBlock Text="Each button accesses
one fixed product directly. The number of lookup steps
stays constant."
                       TextWrapping="Wrap"
                       Foreground="#78465C"/>
        </StackPanel>
    </Border>

    <TextBlock Text="Made with ♡ and a little
sparkle"
               HorizontalAlignment="Center"
               Foreground="#78465C"/>
  </StackPanel>
 </ScrollViewer>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;

namespace BeautyLab01;

public partial class MainWindow : Window
{
    public MainWindow()
    {
        InitializeComponent();
    }

    private void Lipstick_Click(object? sender,
RoutedEventArgs e)
    {
        ResultText.Text = " 💄 You selected Lipstick!";
    }

    private void Foundation_Click(object? sender,
RoutedEventArgs e)
    {
        ResultText.Text = "✨ You selected Foundation!";
    }
```

```
    private void Blush_Click(object? sender,
RoutedEventArgs e)
    {
        ResultText.Text = "🌸 You selected Blush!";
    }

    private void Mascara_Click(object? sender,
RoutedEventArgs e)
    {
        ResultText.Text = "🖤 You selected Mascara!";
    }

    private void LipGloss_Click(object? sender,
RoutedEventArgs e)
    {
        ResultText.Text = "💋 You selected Lip Gloss!";
    }
}
```

Project 02: Beauty Price List (O(1))

1. BeautyLab02.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<BuiltInComInteropSupport>true</BuiltInComInteropS
upport>

<ApplicationManifest>app.manifest</ApplicationManif
est>

<AvaloniaUseCompiledBindingsByDefault>false</Ava
loniaUseCompiledBindingsByDefault>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>

  <ItemGroup>
    <AvaloniaResource Include="Images/**" />
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="BeautyLab02.MainWindow"
        Title="Beauty Corner"
        Width="900"
        Height="700"
        MinWidth="750"
        MinHeight="600"
        Background="#FFF6C9"
        WindowStartupLocation="CenterScreen">

    <Grid ColumnDefinitions="1.1*,1.4*" Margin="22">

        <Border Grid.Column="0"
                Background="#FFD9E8"
                CornerRadius="22"
                Padding="18"
                Margin="0,0,10,0">

            <ScrollViewer>
                <StackPanel Spacing="14">

                    <TextBlock Text="♡ BEAUTY CORNER"
                               FontSize="23"
                               FontWeight="Bold"
                               Foreground="#78465C"/>

                    <TextBlock Text="Choose your makeup!"
                               FontSize="14"
                               Foreground="#78465C"/>

                    <Border Background="#FFFDF5"
CornerRadius="14" Padding="10">
                        <Grid ColumnDefinitions="75,*"
ColumnSpacing="10">
                            <Image
Source="/Images/lipstick.jpeg" Height="75"
Stretch="Uniform"/>
                            <StackPanel Grid.Column="1"
VerticalAlignment="Center" Spacing="8">
                                <TextBlock Text="Lipstick"
FontSize="16" FontWeight="Bold"
Foreground="#78465C"/>
                                <Button Content="View Price ♡"
Background="#F4A7C5" Foreground="#60384B"
Click="Lipstick_Click"/>
                            </StackPanel>
                        </Grid>
                    </Border>

                    <Border Background="#FFFDF5"
CornerRadius="14" Padding="10">
                        <Grid ColumnDefinitions="75,*"
ColumnSpacing="10">
                            <Image
Source="/Images/foundation.jpeg" Height="75"
Stretch="Uniform"/>
                            <StackPanel Grid.Column="1"
VerticalAlignment="Center" Spacing="8">
                                <TextBlock Text="Foundation"
FontSize="16" FontWeight="Bold"
Foreground="#78465C"/>
                                <Button Content="View Price ♡"
Background="#F4A7C5" Foreground="#60384B"
Click="Foundation_Click"/>
```

```
                    </StackPanel>
                </Grid>
            </Border>

            <Border Background="#FFFDF5"
CornerRadius="14" Padding="10">
                <Grid ColumnDefinitions="75,*"
ColumnSpacing="10">
                    <Image
Source="/Images/blush.jpeg" Height="75"
Stretch="Uniform"/>
                    <StackPanel Grid.Column="1"
VerticalAlignment="Center" Spacing="8">
                        <TextBlock Text="Blush"
FontSize="16" FontWeight="Bold"
Foreground="#78465C"/>
                        <Button Content="View Price ♡"
Background="#F4A7C5" Foreground="#60384B"
Click="Blush_Click"/>
                    </StackPanel>
                </Grid>
            </Border>

            <Border Background="#FFFDF5"
CornerRadius="14" Padding="10">
                <Grid ColumnDefinitions="75,*"
ColumnSpacing="10">
                    <Image
Source="/Images/mascara.jpeg" Height="75"
Stretch="Uniform"/>
                    <StackPanel Grid.Column="1"
VerticalAlignment="Center" Spacing="8">
                        <TextBlock Text="Mascara"
FontSize="16" FontWeight="Bold"
Foreground="#78465C"/>
                        <Button Content="View Price ♡"
Background="#F4A7C5" Foreground="#60384B"
Click="Mascara_Click"/>
                    </StackPanel>
                </Grid>
            </Border>

            <Border Background="#FFFDF5"
CornerRadius="14" Padding="10">
                <Grid ColumnDefinitions="75,*"
ColumnSpacing="10">
                    <Image
Source="/Images/lipgloss.jpeg" Height="75"
Stretch="Uniform"/>
                    <StackPanel Grid.Column="1"
VerticalAlignment="Center" Spacing="8">
                        <TextBlock Text="Lip Gloss"
FontSize="16" FontWeight="Bold"
Foreground="#78465C"/>
                        <Button Content="View Price ♡"
Background="#F4A7C5" Foreground="#60384B"
Click="LipGloss_Click"/>
                    </StackPanel>
                </Grid>
            </Border>

        </StackPanel>
    </ScrollViewer>
</Border>

<Border Grid.Column="1"
        Background="#FFFDF5"
        CornerRadius="22"
        Padding="24"
        Margin="10,0,0,0">

    <StackPanel VerticalAlignment="Center"
Spacing="20">

        <TextBlock Text="♡ YOUR PRICE CHECK"
                   FontSize="22"
                   FontWeight="Bold"
                   Foreground="#78465C"
                   HorizontalAlignment="Center"/>

        <Border Background="#FFEAB0"
CornerRadius="20" Padding="20">
            <StackPanel Spacing="14"
HorizontalAlignment="Center">
                <TextBlock Text="✨" FontSize="45"
HorizontalAlignment="Center"/>

                <TextBlock x:Name="ResultText"
                           Text="Choose a product from
the list to see its price!"
                           FontSize="22"
                           FontWeight="Bold"
                           Foreground="#78465C"
                           TextAlignment="Center"
                           TextWrapping="Wrap"/>
            </StackPanel>
        </Border>

        <TextBlock Text="HOW IT WORKS"
                   FontSize="17"
                   FontWeight="Bold"
                   Foreground="#78465C"/>

        <TextBlock Text="Each button directly
displays one fixed product price. No searching
through the product list is needed."
                   FontSize="15"
                   Foreground="#78465C"
                   TextWrapping="Wrap"/>

        <Border Background="#FFD9E8"
CornerRadius="14" Padding="15">
            <StackPanel Spacing="6">
                <TextBlock Text="BIG-O NOTATION"
                           FontSize="14"
                           FontWeight="Bold"
                           Foreground="#78465C"
                           HorizontalAlignment="Center"/>
                <TextBlock Text="O(1) — CONSTANT
TIME"
                           FontSize="20"
                           FontWeight="Bold"
                           Foreground="#B75E85"
                           HorizontalAlignment="Center"/>
```

```
            </StackPanel>
        </Border>

        <TextBlock Text="Made with ♡ and a little
sparkle"
                   FontSize="13"
                   Foreground="#78465C"
                   HorizontalAlignment="Center"/>

    </StackPanel>
  </Border>

</Grid>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;

namespace BeautyLab02
{
  public partial class MainWindow : Window
  {
    public MainWindow()
    {
      InitializeComponent();
    }

    private void Lipstick_Click(object? sender,
RoutedEventArgs e)
    {
      ResultText.Text = "💄 Lipstick\nPrice: ₱250";
    }

    private void Foundation_Click(object? sender,
RoutedEventArgs e)
    {
      ResultText.Text = "✨ Foundation\nPrice:
₱450";
    }

    private void Blush_Click(object? sender,
RoutedEventArgs e)
    {
      ResultText.Text = "🌸 Blush\nPrice: ₱300";
    }

    private void Mascara_Click(object? sender,
RoutedEventArgs e)
    {
      ResultText.Text = "🖤 Mascara\nPrice: ₱280";
    }

    private void LipGloss_Click(object? sender,
RoutedEventArgs e)
    {
      ResultText.Text = "💋 Lip Gloss\nPrice: ₱220";
    }
  }
}
```

Project 03: Apply a Fixed Discount (O(1))

1. BeautyLab03.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<BuiltInComInteropSupport>true</BuiltInComInteropS
upport>

<ApplicationManifest>app.manifest</ApplicationManif
est>

<AvaloniaUseCompiledBindingsByDefault>false</Ava
loniaUseCompiledBindingsByDefault>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>

  <ItemGroup>
    <AvaloniaResource Include="Images/**" />
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="BeautyLab03.MainWindow"
        Title="BeautyLab ♡ Fixed Discount"
        Width="950"
        Height="720"
        Background="#FFF6C9"
        WindowStartupLocation="CenterScreen">

    <ScrollViewer>
      <StackPanel Margin="28" Spacing="20">

        <Border Background="#FFD9E8"
                CornerRadius="24"
                Padding="22">
          <StackPanel Spacing="8">
            <TextBlock Text="♡ B E A U T Y L A B ♡"
                       FontSize="16"
                       FontWeight="Bold"
                       Foreground="#78465C"
                       HorizontalAlignment="Center"/>

            <TextBlock Text="Apply a Fixed
Discount"
                       FontSize="30"
```

```
                    FontWeight="Bold"
                    Foreground="#78465C"
                    HorizontalAlignment="Center"/>

            <TextBlock Text="Save a little on your
beauty favorites! ✨"
                    FontSize="15"
                    Foreground="#78465C"
                    HorizontalAlignment="Center"/>
        </StackPanel>
    </Border>

    <Grid ColumnDefinitions="*,*"
          ColumnSpacing="20">

        <Border Grid.Column="0"
                Background="#FFFDF5"
                CornerRadius="20"
                Padding="20">
            <StackPanel Spacing="14">
                <Image Source="/Images/lipstick.jpeg"
                    Height="280"
                    Stretch="Uniform"/>

                <TextBlock Text="♡ Pink Velvet
Lipstick"
                        FontSize="22"
                        FontWeight="Bold"
                        Foreground="#78465C"
                        HorizontalAlignment="Center"/>

                <TextBlock Text="Original Price:
₱250.00"
                        FontSize="19"
                        Foreground="#78465C"
                        HorizontalAlignment="Center"/>

                <Button Content="Apply 10% Discount
♡"
                    Background="#F4A7C5"
                    Foreground="#60384B"
                    FontSize="17"
                    Padding="12"
                    Click="ApplyDiscount_Click"/>
            </StackPanel>
        </Border>

        <Border Grid.Column="1"
                Background="#FFEAB0"
                CornerRadius="20"
                Padding="24">
            <StackPanel VerticalAlignment="Center"
                    Spacing="20">

                <TextBlock Text="♡ YOUR SAVINGS"
                        FontSize="24"
                        FontWeight="Bold"
                        Foreground="#78465C"
                        HorizontalAlignment="Center"/>

                <TextBlock Text="Discount Rate: 10%"
                        FontSize="18"
                        Foreground="#78465C"
                        HorizontalAlignment="Center"/>

                <TextBlock x:Name="DiscountText"
                        Text="Discount Amount: ₱0.00"
                        FontSize="19"
                        FontWeight="Bold"
                        Foreground="#B75E85"
                        TextAlignment="Center"
                        TextWrapping="Wrap"/>

                <TextBlock x:Name="FinalPriceText"
                        Text="Final Price: ₱250.00"
                        FontSize="24"
                        FontWeight="Bold"
                        Foreground="#78465C"
                        TextAlignment="Center"
                        TextWrapping="Wrap"/>

                <TextBlock x:Name="ResultText"
                        Text="Click the button to
calculate your savings!"
                        FontSize="16"
                        Foreground="#78465C"
                        TextAlignment="Center"
                        TextWrapping="Wrap"/>

                <Border Background="#FFD9E8"
                        CornerRadius="14"
                        Padding="15">
                    <StackPanel Spacing="8">
                        <TextBlock Text="BIG-O
NOTATION"
                                FontSize="14"
                                FontWeight="Bold"
                                Foreground="#78465C"

HorizontalAlignment="Center"/>

                        <TextBlock Text="O(1) —
CONSTANT TIME"
                                FontSize="19"
                                FontWeight="Bold"
                                Foreground="#B75E85"

HorizontalAlignment="Center"/>

                        <TextBlock Text="The discount
uses a fixed number of calculations, regardless of
how many products exist."
                                FontSize="14"
                                Foreground="#78465C"
                                TextWrapping="Wrap"/>
                    </StackPanel>
                </Border>
            </StackPanel>
        </Border>
    </Grid>

    <TextBlock Text="Made with ♡ and a little
sparkle"
            HorizontalAlignment="Center"
```

```
                Foreground="#78465C"/>
        </StackPanel>
    </ScrollViewer>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;

namespace BeautyLab03;

public partial class MainWindow : Window
{
    public MainWindow()
    {
        InitializeComponent();
    }

    private void ApplyDiscount_Click(object? sender,
RoutedEventArgs e)
    {
        decimal originalPrice = 250m;
        decimal discountRate = 0.10m;

        decimal discountAmount = originalPrice *
discountRate;
        decimal finalPrice = originalPrice -
discountAmount;

        DiscountText.Text =
            $"Discount Amount: ₱{discountAmount:F2}";

        FinalPriceText.Text =
            $"Final Price: ₱{finalPrice:F2}";

        ResultText.Text =
            $"Discount applied! You saved
₱{discountAmount:F2}. ♡";
    }
}
```

Project 04: Ice Cream Flavor Search (O(log n))

1. IceCream04.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<BuiltInComInteropSupport>true</BuiltInComInteropS
upport>

<ApplicationManifest>app.manifest</ApplicationManif
est>

<AvaloniaUseCompiledBindingsByDefault>false</Ava
loniaUseCompiledBindingsByDefault>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="IceCream04.MainWindow"
        Title="Scoop &amp;amp; Search"
        Width="850"
        Height="700"
        Background="#FFF5E6"
        WindowStartupLocation="CenterScreen">

    <ScrollViewer>
        <StackPanel Margin="30" Spacing="20">

            <Border Background="#FFD6E7"
                    CornerRadius="25"
                    Padding="25">
                <StackPanel Spacing="10">
                    <TextBlock Text="🍦 SCOOP &amp;amp;
SEARCH 🍦"
                               FontSize="18"
                               FontWeight="Bold"
                               Foreground="#75465E"
                               HorizontalAlignment="Center"/>

                    <TextBlock Text="Ice Cream Flavor
Finder"
                               FontSize="30"
                               FontWeight="Bold"
                               Foreground="#75465E"
                               HorizontalAlignment="Center"/>

                    <TextBlock Text="Find your favorite
scoop!"
                               FontSize="16"
                               Foreground="#75465E"
                               HorizontalAlignment="Center"/>
                </StackPanel>
            </Border>

            <Border Background="#FFFFFF"
                    CornerRadius="20"
                    Padding="22">
                <StackPanel Spacing="15">

                    <TextBlock Text="OUR FLAVORS"
                               FontSize="22"
                               FontWeight="Bold"
                               Foreground="#75465E"
                               HorizontalAlignment="Center"/>
```

```
            <TextBlock Text="Chocolate • Mango •
Matcha • Strawberry • Vanilla"
                       FontSize="16"
                       Foreground="#75465E"
                       TextWrapping="Wrap"
                       HorizontalAlignment="Center"/>

            <TextBlock Text="Enter a flavor to
search:"
                       FontSize="16"
                       Foreground="#75465E"/>

            <TextBox x:Name="SearchInput"
                     Watermark="Example: Mango"
                     FontSize="18"
                     Padding="12"/>

            <Button Content="Find My Flavor ♡"
                    Background="#F4A6C1"
                    Foreground="#60384B"
                    FontSize="18"
                    Padding="14"
                    Click="Search_Click"/>

            <Border Background="#FFF0B8"
                    CornerRadius="15"
                    Padding="18">
              <StackPanel Spacing="12">
                <TextBlock Text="SEARCH
RESULT"
                           FontSize="16"
                           FontWeight="Bold"
                           Foreground="#75465E"

HorizontalAlignment="Center"/>

                <TextBlock x:Name="ResultText"
                           Text="Your result will appear
here! 🍨"
                           FontSize="20"
                           FontWeight="Bold"
                           Foreground="#75465E"
                           TextWrapping="Wrap"

HorizontalAlignment="Center"/>

                <TextBlock x:Name="StepsText"
                           Text="Search steps: 0"
                           FontSize="16"
                           Foreground="#75465E"

HorizontalAlignment="Center"/>
              </StackPanel>
            </Border>
          </StackPanel>
        </Border>

        <Border Background="#D8EEDB"
                CornerRadius="18"
                Padding="18">
          <StackPanel Spacing="8">
            <TextBlock Text="HOW IT WORKS"
                       FontSize="18"
                       FontWeight="Bold"
                       Foreground="#426B52"
                       HorizontalAlignment="Center"/>

            <TextBlock Text="Binary Search checks
the middle flavor first, then searches only the half
where your flavor could be."
                       FontSize="15"
                       Foreground="#426B52"
                       TextWrapping="Wrap"
                       HorizontalAlignment="Center"/>

            <TextBlock Text="Time Complexity: O(log
n)"
                       FontSize="18"
                       FontWeight="Bold"
                       Foreground="#426B52"
                       HorizontalAlignment="Center"/>
          </StackPanel>
        </Border>

        <TextBlock Text="Made with love and ice
cream ♡"
                   Foreground="#75465E"
                   HorizontalAlignment="Center"/>

      </StackPanel>
    </ScrollViewer>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;
using System;

namespace IceCream04
{
    public partial class MainWindow : Window
    {
        public MainWindow()
        {
            InitializeComponent();
        }

        private void Search_Click(object? sender,
RoutedEventArgs e)
        {
            string[] flavors =
            {
                "Chocolate",
                "Mango",
                "Matcha",
                "Strawberry",
                "Vanilla"
            };

            string target = SearchInput.Text?.Trim() ?? "";

            if (string.IsNullOrWhiteSpace(target))
            {
                ResultText.Text = "Please enter an ice
cream flavor! 🍦";
```

```
            StepsText.Text = "Search steps: 0";
            return;
        }

        int left = 0;
        int right = flavors.Length - 1;
        int steps = 0;

        while (left <= right)
        {
            steps++;

            int middle = left + (right - left) / 2;

            int comparison = string.Compare(
                flavors[middle],
                target,
                StringComparison.OrdinalIgnoreCase
            );

            if (comparison == 0)
            {
                ResultText.Text = $"🍦 Found
{flavors[middle]}!";
                StepsText.Text = $"Search steps:
{steps}";
                return;
            }
            else if (comparison < 0)
            {
                left = middle + 1;
            }
            else
            {
                right = middle - 1;
            }
        }

        ResultText.Text = $"Sorry! {target} is not on
our menu. 🍨";
        StepsText.Text = $"Search steps: {steps}";
    }
  }
}
```

Project 05: Ice Cream Price Lookup (O(log n))

1. IceCream05.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<BuiltInComInteropSupport>true</BuiltInComInteropS
upport>

<ApplicationManifest>app.manifest</ApplicationManif
est>

<AvaloniaUseCompiledBindingsByDefault>false</Ava
loniaUseCompiledBindingsByDefault>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="IceCream05.MainWindow"
        Title="Scoop &amp;amp; Save"
        Width="950"
        Height="720"
        Background="#FFF6C9"
        WindowStartupLocation="CenterScreen">

    <Grid ColumnDefinitions="*,*"
          RowDefinitions="Auto,*"
          Margin="25">

        <Border Grid.ColumnSpan="2"
                Background="#FF8FB1"
                CornerRadius="20"
                Padding="20"
                Margin="0,0,0,20">
            <StackPanel>
                <TextBlock Text="SCOOP &amp;amp; SAVE"
                           FontSize="30"
                           FontWeight="Bold"
                           Foreground="White"
                           HorizontalAlignment="Center"/>
                <TextBlock Text="Find the price of your
favorite flavor!"
                           FontSize="16"
                           Foreground="White"
                           HorizontalAlignment="Center"/>
            </StackPanel>
        </Border>

        <Border Grid.Row="1"
                Grid.Column="0"
                Background="White"
                CornerRadius="20"
                Padding="20"
                Margin="0,0,10,0">
            <StackPanel Spacing="15">
                <TextBlock Text="🍦 Price Lookup"
                           FontSize="24"
                           FontWeight="Bold"
                           Foreground="#9B4D72"/>
```

```
            <TextBlock Text="Available flavors:"
                       FontSize="16"
                       FontWeight="Bold"/>

            <TextBlock Text="Chocolate -
₱80&#10;Mango - ₱70&#10;Matcha -
₱90&#10;Strawberry - ₱75&#10;Vanilla - ₱60"
                       FontSize="18"
                       LineHeight="32"/>

            <TextBox x:Name="SearchInput"
                     PlaceholderText="Enter a flavor..."
                     FontSize="16"
                     Padding="10"/>

            <Button Content="Find Price"
                    Background="#FF8FB1"
                    Foreground="White"
                    FontSize="16"
                    Padding="12"
                    CornerRadius="10"
                    Click="Search_Click"/>
        </StackPanel>
    </Border>

    <Border Grid.Row="1"
            Grid.Column="1"
            Background="#FFE0EB"
            CornerRadius="20"
            Padding="20"
            Margin="10,0,0,0">
        <StackPanel Spacing="15">
            <TextBlock Text="Your Result"
                       FontSize="24"
                       FontWeight="Bold"
                       Foreground="#9B4D72"/>

            <TextBlock x:Name="ResultText"
                       Text="The price will appear here."
                       FontSize="20"
                       TextWrapping="Wrap"/>

            <TextBlock x:Name="StepsText"
                       Text="Search steps: 0"
                       FontSize="16"
                       FontWeight="Bold"/>

            <TextBlock Text="About the Algorithm"
                       FontSize="20"
                       FontWeight="Bold"
                       Foreground="#9B4D72"
                       Margin="0,15,0,0"/>

            <TextBlock Text="Binary Search checks the
middle item and removes half of the remaining
choices after each comparison."
                       FontSize="16"
                       TextWrapping="Wrap"/>

            <TextBlock Text="Time Complexity: O(log
n)"
                       FontSize="18"
                       FontWeight="Bold"
                       Foreground="#9B4D72"/>
        </StackPanel>
    </Border>

</Grid>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;
using System;

namespace IceCream05
{
    public partial class MainWindow : Window
    {
        public MainWindow()
        {
            InitializeComponent();
        }

        private void Search_Click(object? sender,
RoutedEventArgs e)
        {
            string[] flavors =
            {
                "Chocolate",
                "Mango",
                "Matcha",
                "Strawberry",
                "Vanilla"
            };

            decimal[] prices = { 80, 70, 90, 75, 60 };

            string target = SearchInput.Text?.Trim() ?? "";

            if (string.IsNullOrWhiteSpace(target))
            {
                ResultText.Text = "Please enter an ice
cream flavor!";
                StepsText.Text = "Search steps: 0";
                return;
            }

            int left = 0;
            int right = flavors.Length - 1;
            int steps = 0;

            while (left <= right)
            {
                steps++;

                int middle = left + (right - left) / 2;

                int comparison = string.Compare(
                    flavors[middle],
                    target,
                    StringComparison.OrdinalIgnoreCase
                );
```

```
            if (comparison == 0)
            {
                ResultText.Text =
                    $"{flavors[middle]} costs
₱{prices[middle]:0.00}!";
                StepsText.Text = $"Search steps:
{steps}";
                return;
            }
            else if (comparison < 0)
            {
                left = middle + 1;
            }
            else
            {
                right = middle - 1;
            }
        }

        ResultText.Text = $"{target} is not on our
menu.";
        StepsText.Text = $"Search steps: {steps}";
    }
  }
}
```

Project 06: Ice Cream Order ID Search (O(log n))

1. IceCream06.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<BuiltInComInteropSupport>true</BuiltInComInteropS
upport>

<ApplicationManifest>app.manifest</ApplicationManif
est>

<AvaloniaUseCompiledBindingsByDefault>false</Ava
loniaUseCompiledBindingsByDefault>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
    x:Class="IceCream06.MainWindow"
    Title="Ice Cream Order ID Search"
    Width="900"
    Height="650"
    Background="Red"
    WindowStartupLocation="CenterScreen">

  <ScrollViewer>
    <StackPanel Margin="30" Spacing="20">

      <TextBlock Text="ICE CREAM ORDER
SEARCH"
                 FontSize="30"
                 FontWeight="Bold"
                 Foreground="White"
                 HorizontalAlignment="Center"
                 TextWrapping="Wrap"/>

      <Border Background="White"
              Padding="25"
              CornerRadius="15">
        <StackPanel Spacing="15">

          <TextBlock Text="Available Orders"
                     FontSize="24"
                     FontWeight="Bold"
                     Foreground="DarkRed"/>

          <TextBlock Text="1001 -
Vanilla&#10;1002 - Chocolate&#10;1003 -
Strawberry&#10;1004 - Cookies and
Cream&#10;1005 - Mango"
                     FontSize="18"
                     Foreground="Black"
                     TextWrapping="Wrap"/>

          <TextBlock Text="Enter Order ID:"
                     FontSize="18"
                     FontWeight="Bold"
                     Foreground="Black"/>

          <TextBox x:Name="OrderIdInput"
                   PlaceholderText="Example: 1003"
                   FontSize="18"
                   Padding="12"/>

          <Button Content="Search Order"
                  Background="DarkRed"
                  Foreground="White"
                  FontSize="18"
                  Padding="12"
                  Click="Search_Click"/>

        </StackPanel>
      </Border>

      <Border Background="White"
              Padding="25"
              CornerRadius="15">
        <StackPanel Spacing="12">

          <TextBlock Text="Search Result"
```

```
                    FontSize="24"
                    FontWeight="Bold"
                    Foreground="DarkRed"/>

            <TextBlock x:Name="ResultText"
                    Text="Your search result will
appear here."
                    FontSize="20"
                    Foreground="Black"
                    TextWrapping="Wrap"/>

            <TextBlock x:Name="StepsText"
                    Text="Steps: 0"
                    FontSize="18"
                    Foreground="Black"/>

            <TextBlock Text="Algorithm: Binary
Search"
                    FontSize="20"
                    FontWeight="Bold"
                    Foreground="DarkRed"/>

            <TextBlock Text="Binary Search checks
the middle order ID and repeatedly halves the search
range."
                    FontSize="17"
                    Foreground="Black"
                    TextWrapping="Wrap"/>

            <TextBlock Text="Time Complexity: O(log
n)"
                    FontSize="20"
                    FontWeight="Bold"
                    Foreground="DarkRed"/>

        </StackPanel>
      </Border>

    </StackPanel>
  </ScrollViewer>
</Window>
```

3. MainWindow.axaml..cs

```
<![CDATA[using Avalonia.Controls; using
Avalonia.Interactivity; namespace IceCream06 {
public partial class MainWindow : Window { public
MainWindow() { InitializeComponent(); } private void
Search_Click(object? sender, RoutedEventArgs e) {
int[] orderIds = { 1001, 1002, 1003, 1004, 1005 };
string[] iceCreamNames = { "Vanilla", "Chocolate",
"Strawberry", "Cookies and Cream", "Mango" }; if
(!int.TryParse(OrderIdInput.Text, out int targetId)) {
ResultText.Text = "Please enter a valid numeric Order
ID."; StepsText.Text = "Steps: 0"; return; } int left = 0;
int right = orderIds.Length - 1; int steps = 0; bool
found = false; while (left <= right) { steps++; int middle
= left + (right - left) / 2; if (orderIds[middle] == targetId)
{ ResultText.Text = "Order Found!\nOrder ID: " +
orderIds[middle] + "\nIce Cream: " +
iceCreamNames[middle]; found = true; break; } else if
(orderIds[middle] < targetId) { left = middle + 1; } else {
right = middle - 1; } } if (!found) { ResultText.Text =
"Order ID " + targetId + " was not found."; }
StepsText.Text = "Steps: " + steps; } } } ]]>
```

Project 07: Color Finder (O(n))

1. ***Color07.csproj***

```
<Project Sdk="Microsoft.NET.Sdk">

  <PropertyGroup>

    <OutputType>WinExe</OutputType>

    <TargetFramework>net10.0</TargetFramework>

    <Nullable>enable</Nullable>


<BuiltInComInteropSupport>true</BuiltInComInteropS
upport>


<ApplicationManifest>app.manifest</ApplicationManif
est>


<AvaloniaUseCompiledBindingsByDefault>false</Ava
loniaUseCompiledBindingsByDefault>

  </PropertyGroup>



  <ItemGroup>

    <PackageReference Include="Avalonia"
Version="12.1.4" />

    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />

    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>

    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />

  </ItemGroup>

</Project>
```

2. ***MainWindow.axaml***

```
<Window xmlns="https://github.com/avaloniaui"


xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
```

```
    x:Class="Color07.MainWindow"
    Title="Color Finder"
    Width="900"
    Height="650"
    Background="Red"
    WindowStartupLocation="CenterScreen">

<ScrollViewer>
    <StackPanel Margin="30" Spacing="20">

        <TextBlock Text="COLOR FINDER"
                   FontSize="36"
                   FontWeight="Bold"
                   Foreground="White"
                   HorizontalAlignment="Center"/>

        <Border Background="White"
                Padding="25"
                CornerRadius="15">
            <StackPanel Spacing="15">
                <TextBlock Text="Available Colors"
                           FontSize="25"
                           FontWeight="Bold"
                           Foreground="DarkRed"/>

                <TextBlock Text="Red | Blue | Green |
Purple | Yellow"
                           FontSize="20"
                           FontWeight="Bold"
                           Foreground="Black"
                           TextWrapping="Wrap"/>

                <TextBlock Text="Enter a color to
search:"
                           FontSize="18"
                           Foreground="Black"/>

                <TextBox x:Name="SearchInput"
                         FontSize="18"
                         Padding="12"
                         PlaceholderText="Example:
Purple"/>

                <Button Content="Find Color"
                        Background="DarkRed"
                        Foreground="White"
                        FontSize="20"
                        Padding="12"
                        Click="Search_Click"/>
            </StackPanel>
        </Border>

        <Border Background="White"
                Padding="25"
                CornerRadius="15">
            <StackPanel Spacing="12">
                <TextBlock Text="Search Result"
                           FontSize="25"
                           FontWeight="Bold"
                           Foreground="DarkRed"/>
```

```
            <TextBlock x:Name="ResultText"
                       Text="Your result will appear
here."
                       FontSize="20"
                       Foreground="Black"
                       TextWrapping="Wrap"/>

            <TextBlock x:Name="StepsText"
                       Text="Colors checked: 0"
                       FontSize="18"
                       Foreground="Black"/>

            <TextBlock Text="Algorithm: foreach
loop"
                       FontSize="20"
                       FontWeight="Bold"
                       Foreground="DarkRed"/>

            <TextBlock Text="The program checks
each color until it finds a match."
                       FontSize="17"
                       Foreground="Black"
                       TextWrapping="Wrap"/>

            <TextBlock Text="Time Complexity: O(n)"
                       FontSize="20"
                       FontWeight="Bold"
                       Foreground="DarkRed"/>
        </StackPanel>
    </Border>
    </StackPanel>
  </ScrollViewer>
</Window>
```

3. ***MainWindow.axaml.cs***

```
<![CDATA[using Avalonia.Controls;
using Avalonia.Interactivity;
using System;

namespace Color07
{
  public partial class MainWindow : Window
  {
    public MainWindow()
    {
      InitializeComponent();
    }

    private void Search_Click(object? sender,
RoutedEventArgs e)
    {
      string[] colors =
      {
        "Red",
        "Blue",
        "Green",
        "Purple",
        "Yellow"
      };

      string target = SearchInput.Text?.Trim() ?? "";
      int checkedColors = 0;
      bool found = false;

      if (string.IsNullOrWhiteSpace(target))
      {
        ResultText.Text = "Please enter a color!";
        StepsText.Text = "Colors checked: 0";
        return;
      }

      foreach (string color in colors)
      {
        checkedColors++;

        if (string.Equals(
          color,
          target,
          StringComparison.OrdinalIgnoreCase))
        {
          ResultText.Text = "Found the color " +
color + "!";
          found = true;
          break;
```

```
            }
        }

        if (!found)
        {
            ResultText.Text =
                target + " is not in the color list.";
        }

        StepsText.Text = "Colors checked: " +
checkedColors;
    }
  }
}
]]>
```

Project 08: Color Palette Finder (O(n))

1. ***Color08.csproj***

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<BuiltInComInteropSupport>true</BuiltInComInteropS
upport>

<ApplicationManifest>app.manifest</ApplicationManif
est>

<AvaloniaUseCompiledBindingsByDefault>false</Ava
loniaUseCompiledBindingsByDefault>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>
</Project>
```

2. ***MainWindow.axaml***

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="Color08.MainWindow"
        Title="Chromatic | Color Palette Finder"
        Width="720"
        Height="650"
        MinWidth="650"
        MinHeight="580"
        Background="#111827"
        WindowStartupLocation="CenterScreen">

    <StackPanel Margin="20" Spacing="12">
        <StackPanel Spacing="4">
            <TextBlock Text="CHROMATIC"
                       FontSize="12"
                       FontWeight="Bold"
                       Foreground="#A78BFA"
                       HorizontalAlignment="Center"/>

            <TextBlock Text="Color Palette Finder"
                       FontSize="28"
                       FontWeight="Bold"
                       Foreground="White"
                       HorizontalAlignment="Center"/>

            <TextBlock Text="Explore colors with Binary
Search"
                       FontSize="14"
                       Foreground="#D1D5DB"
                       HorizontalAlignment="Center"/>
        </StackPanel>

        <Border Background="White"
CornerRadius="16" Padding="16">
            <StackPanel Spacing="10">
                <TextBlock Text="YOUR COLOR
COLLECTION"
                           FontSize="13"
                           FontWeight="Bold"
                           Foreground="#6D28D9"/>

                <Grid ColumnDefinitions="*,*,*"
RowDefinitions="Auto,Auto"
                      ColumnSpacing="8" RowSpacing="8">

                    <Border Grid.Row="0" Grid.Column="0"
Background="#EF4444"
                            CornerRadius="8" Padding="10">
                        <StackPanel Spacing="3">
                            <TextBlock Text="RED"
Foreground="White" FontWeight="Bold"/>
                            <TextBlock Text="#EF4444"
Foreground="White" FontSize="12"/>
                        </StackPanel>
                    </Border>

                    <Border Grid.Row="0" Grid.Column="1"
Background="#3B82F6"
                            CornerRadius="8" Padding="10">
                        <StackPanel Spacing="3">
                            <TextBlock Text="BLUE"
Foreground="White" FontWeight="Bold"/>
                            <TextBlock Text="#3B82F6"
Foreground="White" FontSize="12"/>
                        </StackPanel>
                    </Border>

                    <Border Grid.Row="0" Grid.Column="2"
Background="#10B981"
                            CornerRadius="8" Padding="10">
                        <StackPanel Spacing="3">
                            <TextBlock Text="GREEN"
Foreground="White" FontWeight="Bold"/>
```

```
                    <TextBlock Text="#10B981"
Foreground="White" FontSize="12"/>
                </StackPanel>
            </Border>

            <Border Grid.Row="1" Grid.Column="0"
Background="#8B5CF6"
                    CornerRadius="8" Padding="10">
                <StackPanel Spacing="3">
                    <TextBlock Text="PURPLE"
Foreground="White" FontWeight="Bold"/>
                    <TextBlock Text="#8B5CF6"
Foreground="White" FontSize="12"/>
                </StackPanel>
            </Border>

            <Border Grid.Row="1" Grid.Column="1"
Background="#F59E0B"
                    CornerRadius="8" Padding="10">
                <StackPanel Spacing="3">
                    <TextBlock Text="YELLOW"
Foreground="#111827" FontWeight="Bold"/>
                    <TextBlock Text="#F59E0B"
Foreground="#111827" FontSize="12"/>
                </StackPanel>
            </Border>

            <Border Grid.Row="1" Grid.Column="2"
Background="#EC4899"
                    CornerRadius="8" Padding="10">
                <StackPanel Spacing="3">
                    <TextBlock Text="PINK"
Foreground="White" FontWeight="Bold"/>
                    <TextBlock Text="#EC4899"
Foreground="White" FontSize="12"/>
                </StackPanel>
            </Border>

        </Grid>
    </StackPanel>
</Border>

<Border Background="#F3F4F6"
CornerRadius="16" Padding="16">
    <StackPanel Spacing="10">
        <TextBlock Text="FIND A COLOR"
                FontSize="19"
                FontWeight="Bold"
                Foreground="#111827"/>

        <TextBox x:Name="SearchInput"
                PlaceholderText="Enter a color, e.g.
Purple"
                FontSize="16"
                Padding="10"/>

        <Button Content="SEARCH PALETTE"
                Background="#7C3AED"
                Foreground="White"
                FontSize="15"
                FontWeight="Bold"
                Padding="10"
                CornerRadius="8"
                Click="Search_Click"/>
    </StackPanel>
</Border>

<Border Background="#1F2937"
CornerRadius="16" Padding="16">
    <StackPanel Spacing="8">
        <TextBlock Text="SEARCH RESULTS"
                FontSize="12"
                FontWeight="Bold"
                Foreground="#A78BFA"/>

        <TextBlock x:Name="ResultText"
                Text="Your search result will appear
here."
                FontSize="19"
                FontWeight="Bold"
                Foreground="White"
                TextWrapping="Wrap"/>

        <TextBlock x:Name="StepsText"
                Text="Search steps: 0"
                FontSize="14"
                Foreground="#D1D5DB"/>

        <TextBlock Text="BINARY SEARCH  •
TIME COMPLEXITY: O(log n)"
                FontSize="13"
                FontWeight="Bold"
                Foreground="#86EFAC"
                TextWrapping="Wrap"/>
    </StackPanel>
</Border>

<TextBlock Text="CHROMATIC • COLOR
EXPLORER"
            FontSize="10"
            Foreground="#9CA3AF"
            HorizontalAlignment="Center"/>

</StackPanel>
</Window>
```

3. ***MainWindow.axaml.cs***

```
using Avalonia.Controls;
using Avalonia.Interactivity;
using System;

namespace Color08
{
    public partial class MainWindow : Window
    {
        public MainWindow()
        {
            InitializeComponent();
        }

        private void Search_Click(object?
sender, RoutedEventArgs e)
        {
            string[] colors =
            {
```

```
            "Blue",
            "Green",
            "Pink",
            "Purple",
            "Red",
            "Yellow"
        };

        string target =
SearchInput.Text?.Trim() ?? "";

        if (string.IsNullOrWhiteSpace(target))
        {
            ResultText.Text = "Please enter a
color name.";
            StepsText.Text = "Search steps:
0";
            return;
        }

        int steps = 0;
        bool found = false;

        foreach (string color in colors)
        {
            steps++;

            if (string.Equals(
                color,
                target,

StringComparison.OrdinalIgnoreCase))
            {
                ResultText.Text = "Color Found!
" + color;
                found = true;
                break;
            }
        }

        if (!found)
        {
            ResultText.Text = target + " was
not found in the palette.";
        }

        StepsText.Text = "Search steps: " +
steps;
    }
  }
}
```

Project 09: Ice Cream Flavor Search (O(n))

1. ***IceCream.csproj***

```
<Project Sdk="Microsoft.NET.Sdk">

  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>
<ApplicationManifest>app.manifest</ApplicationManif
est>

<AvaloniaUseCompiledBindingsByDefault>true</Aval
oniaUseCompiledBindingsByDefault>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>

  <ItemGroup Condition="'$(Configuration)' ==
'Debug'">
    <PackageReference
Include="Avalonia.Diagnostics" Version="12.1.4" />
  </ItemGroup>

</Project>
```

2. ***MainWindow.axaml***

```
        <Window
        xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"

xmlns:d="http://schemas.microsoft.com/expression/bl
end/2008"

xmlns:mc="http://schemas.openxmlformats.org/marku
p-compatibility/2006"
        mc:Ignorable="d"
        d:DesignWidth="800"
        d:DesignHeight="450"
        x:Class="IceCream09.MainWindow"
        Title="Ice Cream Flavor Search"
        Width="450"
        Height="350">

    <StackPanel Margin="20" Spacing="12">
        <TextBlock Text="Ice Cream Flavor Search
(O(n))"
                FontSize="22"
                FontWeight="Bold"
                TextWrapping="Wrap"/>

        <TextBlock Text="Available flavors: Vanilla,
Strawberry, Chocolate, Matcha, Ube, Mango"
                TextWrapping="Wrap"/>

        <TextBox x:Name="SearchInput"
                PlaceholderText="Enter an ice cream
flavor"/>
```

```
        <Button Content="Search"
                Click="Search_Click"></Button>

        <TextBlock x:Name="ResultText"
                   Text="Enter a flavor to begin."
                   TextWrapping="Wrap"
                   FontSize="16"/>

        <TextBlock x:Name="StepsText"
                   Text="Search steps: 0"/>
    </StackPanel>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;
using System;

namespace IceCream09;

public partial class MainWindow : Window
{
    public MainWindow()
    {
        InitializeComponent();
    }

    private void Search_Click(object? sender,
RoutedEventArgs e)
    {
        string[] flavors =
        {
            "Vanilla",
            "Strawberry",
            "Chocolate",
            "Matcha",
            "Ube",
            "Mango"
        };

        string target = SearchInput.Text?.Trim() ?? "";

        if (string.IsNullOrWhiteSpace(target))
        {
            ResultText.Text = "Please enter an ice cream
flavor.";
            StepsText.Text = "Search steps: 0";
            return;
        }

        int steps = 0;
        bool found = false;

        foreach (string flavor in flavors)
        {
            steps++;

            if (string.Equals(
                flavor,
                target,
                StringComparison.OrdinalIgnoreCase))
            {
                ResultText.Text = "Flavor Found! " + flavor;
                found = true;
                break;
            }
        }

        if (!found)
        {
            ResultText.Text =
                target + " was not found in the flavor list.";
        }

        StepsText.Text = "Search steps: " + steps;
    }
}
```

Project 10: Star Pair Finder

1. Star10.csproj

```
<Project Sdk="Microsoft.NET.Sdk">

  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<ApplicationManifest>app.manifest</ApplicationManif
est>

<AvaloniaUseCompiledBindingsByDefault>true</Aval
oniaUseCompiledBindingsByDefault>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
  </ItemGroup>

  <ItemGroup Condition="'$(Configuration)' ==
'Debug'">
    <PackageReference
Include="Avalonia.Diagnostics" Version="12.1.4" />
  </ItemGroup>

</Project>
```

2. MainWindow.axaml

```
python3 -c 'from pathlib import Path;
Path("MainWindow.axaml").write_text("""<Window
xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
```

```
xmlns:d="http://schemas.microsoft.com/expression/blend/2008"

xmlns:mc="http://schemas.openxmlformats.org/markup-compatibility/2006"
        mc:Ignorable="d"
        x:Class="Star10.MainWindow"
        Title="Star Pair Finder"
        Width="540" Height="700"
        MinWidth="420" MinHeight="550"
        Background="#FFF9E8">

    <ScrollViewer>
        <StackPanel Margin="28" Spacing="18">

            <TextBlock Text="✦  LITTLE STAR STUDIO ✦"
                       Foreground="#72551D"
                       FontSize="13"
                       FontWeight="Bold"
                       HorizontalAlignment="Center"/>

            <TextBlock Text="Star Pair Finder"
                       Foreground="#342B20"
                       FontSize="30"
                       FontWeight="Bold"
                       HorizontalAlignment="Center"/>

            <TextBlock Text="Find every possible pair among your favorite stars."
                       Foreground="#5E5548"
                       FontSize="15"
                       TextWrapping="Wrap"
                       TextAlignment="Center"
                       HorizontalAlignment="Center"/>

            <Border Background="#FFF0B8"
                    BorderBrush="#E6CB73"
                    BorderThickness="1"
                    CornerRadius="18"
                    Padding="20">
                <StackPanel Spacing="14">
                    <TextBlock Text="✧  YOUR STAR COLLECTION"
                               Foreground="#72551D"
                               FontSize="14"
                               FontWeight="Bold"/>

                    <TextBlock Text="⭐  Sirius      ⭐  Vega"
                               Foreground="#342B20"
                               FontSize="18"/>

                    <TextBlock Text="⭐  Polaris      ⭐  Rigel"
                               Foreground="#342B20"
                               FontSize="18"/>

                    <TextBlock Text="⭐  Deneb"
                               Foreground="#342B20"
                               FontSize="18"/>
                </StackPanel>
            </Border>

            <Border Background="#FFFFFF"
                    BorderBrush="#E6DDBF"
                    BorderThickness="1"
                    CornerRadius="18"
                    Padding="20">
                <StackPanel Spacing="12">
                    <TextBlock Text="READY TO EXPLORE?"
                               Foreground="#5E5548"
                               FontSize="13"
                               FontWeight="Bold"/>

                    <Button Content="✦  Find Star Pairs  ✦"
                            Click="FindPairs_Click"
                            Background="#F4D66B"
                            Foreground="#342B20"
                            FontSize="17"
                            FontWeight="Bold"
                            Padding="15"

                            HorizontalContentAlignment="Center"/>
                </StackPanel>
            </Border>

            <Border Background="#FFFFFF"
                    BorderBrush="#E6DDBF"
                    BorderThickness="1"
                    CornerRadius="18"
                    Padding="20">
                <StackPanel Spacing="14">
                    <TextBlock Text="YOUR DISCOVERY RESULTS"
                               Foreground="#72551D"
                               FontSize="14"
                               FontWeight="Bold"/>

                    <TextBlock x:Name="ResultText"
                               Text="Your star pairs will appear here after you click the button."
                               Foreground="#342B20"
                               FontSize="15"
                               TextWrapping="Wrap"/>

                    <Border Background="#FFF0B8"
                            CornerRadius="12"
                            Padding="14">
                        <TextBlock x:Name="StepsText"
                                   Text="Comparisons: 0"
                                   Foreground="#342B20"
                                   FontSize="16"
                                   FontWeight="Bold"/>
                    </Border>
                </StackPanel>
            </Border>

            <TextBlock Text="QUADRATIC TIME COMPLEXITY  •  O(n²)"
                       Foreground="#72551D"
                       FontSize="12"
```

```
                FontWeight="Bold"
                HorizontalAlignment="Center"
                TextAlignment="Center"/>

        <TextBlock Text="Made under a little starlight
✨"
                Foreground="#766B58"
                FontSize="13"
                HorizontalAlignment="Center"/>

      </StackPanel>
   </ScrollViewer>
</Window>
"""')'
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;
using System.Text;

namespace Star10;

public partial class MainWindow : Window
{
   public MainWindow()
   {
      InitializeComponent();
   }

   private void FindPairs_Click(object? sender,
RoutedEventArgs e)
   {
      string[] stars =
      {
         "Sirius",
         "Vega",
         "Polaris",
         "Rigel",
         "Deneb"
      };

      StringBuilder result = new StringBuilder();
      int comparisons = 0;

      for (int i = 0; i < stars.Length; i++)
      {
         for (int j = i + 1; j < stars.Length; j++)
         {
            comparisons++;

            result.AppendLine(
               stars[i] + " + " + stars[j]);
         }
      }

      ResultText.Text = "Star pairs:\n" +
result.ToString();
      StepsText.Text = "Comparisons: " +
comparisons;
   }
}
```

Project 11: Star Pattern Builder (O(n²))

1. Star11.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<ApplicationManifest>app.manifest</ApplicationManif
est>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
    <PackageReference
Include="AvaloniaUI.DiagnosticsSupport"
Version="2.2.3">
      <IncludeAssets Condition="'$(Configuration)' !=
'Debug'">None</IncludeAssets>
      <PrivateAssets Condition="'$(Configuration)' !=
'Debug'">All</PrivateAssets>
    </PackageReference>
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="Star11.MainWindow"
        Title="Star11 - Star Pattern Builder"
        Width="600"
        Height="700"
        MinWidth="400"
        MinHeight="550"
        WindowStartupLocation="CenterScreen"
        Background="#FFF9F0">

    <ScrollViewer>
      <Grid Margin="24">
        <Border Background="#FFFFFF"
                CornerRadius="22"
                Padding="28"
                MaxWidth="440"
                HorizontalAlignment="Center"
                VerticalAlignment="Center">

          <StackPanel Spacing="18">

            <StackPanel Spacing="6"
                        HorizontalAlignment="Center">
```

```
                <TextBlock Text="✦"
                           FontSize="38"
                           Foreground="#D9B84C"
                           HorizontalAlignment="Center"/>

                <TextBlock Text="Star Pattern Builder"
                           FontSize="25"
                           FontWeight="Bold"
                           Foreground="#514333"
                           TextAlignment="Center"
                           TextWrapping="Wrap"/>

                <TextBlock Text="Create your own little
constellation."
                           FontSize="13"
                           Foreground="#958775"
                           TextAlignment="Center"
                           TextWrapping="Wrap"/>
            </StackPanel>

            <Border Height="1"
                    Background="#F0E8DB"/>

            <StackPanel Spacing="8">
              <TextBlock Text="NUMBER OF
ROWS"
                         FontSize="11"
                         FontWeight="Bold"
                         Foreground="#79664D"/>

              <TextBox x:Name="RowsInput"
                       Text="5"

TextChanged="RowsInput_TextChanged"
                       Watermark="Enter a number
from 1 to 10"
                       FontSize="15"
                       Padding="12"
                       Background="#FFFCF7"
                       Foreground="#514333"
                       BorderBrush="#E8DCC9"
                       BorderThickness="1"
                       CornerRadius="10"/>
            </StackPanel>

            <Button x:Name="BuildPatternButton"
                    Content="✦  BUILD STAR
PATTERN  ✦"
                    Background="#F2D779"
                    Foreground="#514333"
                    FontSize="13"
                    FontWeight="Bold"
                    Padding="14"
                    HorizontalAlignment="Stretch"

HorizontalContentAlignment="Center"
                    CornerRadius="12"/>

            <StackPanel Spacing="10">
              <TextBlock Text="YOUR PATTERN"
                         FontSize="11"
                         FontWeight="Bold"
                         Foreground="#79664D"/>

              <Border Background="#FFFCF7"
                      BorderBrush="#F0E8DB"
                      BorderThickness="1"
                      CornerRadius="14"
                      Padding="20"
                      MinHeight="180">

                <TextBlock x:Name="PatternText"
                           Text="Your stars will appear
here."
                           FontFamily="Menlo,
Consolas, monospace"
                           FontSize="18"
                           Foreground="#9A7930"
                           TextWrapping="NoWrap"

HorizontalAlignment="Center"
                           TextAlignment="Center"/>
              </Border>
            </StackPanel>

            <Border Background="#F8F0DF"
                    CornerRadius="10"
                    Padding="14">

              <StackPanel Spacing="5">
                <TextBlock Text="PATTERN
DETAILS"
                           FontSize="10"
                           FontWeight="Bold"
                           Foreground="#8D7756"/>

                <TextBlock x:Name="StepsText"
                           Text="Stars created: 0"
                           FontSize="14"
                           FontWeight="SemiBold"
                           Foreground="#514333"/>
              </StackPanel>
            </Border>

            <TextBlock Text="Made with care and a
little stardust."
                       FontSize="11"
                       Foreground="#B0A18D"
                       HorizontalAlignment="Center"
                       TextAlignment="Center"
                       TextWrapping="Wrap"/>

          </StackPanel>
        </Border>
      </Grid>
    </ScrollViewer>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;
using System.Text;
```

```
namespace Star11;

public partial class MainWindow : Window
{
    public MainWindow()
    {
        InitializeComponent();
    }

    public void BuildPattern_Click(object? sender,
RoutedEventArgs e)
    {
        UpdatePattern();
    }

    public void RowsInput_TextChanged(object?
sender, TextChangedEventArgs e)
    {
        UpdatePattern();
    }

    private void UpdatePattern()
    {
        if (!int.TryParse(RowsInput.Text, out int rows) ||
            rows < 1 || rows > 10)
        {
            PatternText.Text = "Enter a number from 1 to
10.";
            StepsText.Text = "Invalid input";
            return;
        }

        StringBuilder pattern = new StringBuilder();
        int operations = 0;

        for (int i = 1; i <= rows; i++)
        {
            for (int j = 1; j <= i; j++)
            {
                pattern.Append("★ ");
                operations++;
            }

            pattern.AppendLine();
        }

        PatternText.Text = pattern.ToString();
        StepsText.Text = $"Stars created: {operations}";
    }
}
```

Project 12 Star Grid Analyzer (O(n²)).

1. Star12.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<ApplicationManifest>app.manifest</ApplicationManif
est>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
    <PackageReference
Include="AvaloniaUI.DiagnosticsSupport"
Version="2.2.3">
      <IncludeAssets Condition="'$(Configuration)' !=
'Debug'">None</IncludeAssets>
      <PrivateAssets Condition="'$(Configuration)' !=
'Debug'">All</PrivateAssets>
    </PackageReference>
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="Star12.MainWindow"
        Title="Star12 - Star Grid Analyzer"
        Width="360"
        Height="480"
        MinWidth="340"
        MinHeight="440"
        Background="#FFF9E8"
        FontFamily="Arial">

    <StackPanel Margin="18" Spacing="12">

        <TextBlock Text="✦ STAR GRID ANALYZER ✦"
                   FontSize="19"
                   FontWeight="Bold"
                   Foreground="#8A6B16"
                   HorizontalAlignment="Center"
                   TextAlignment="Center"/>

        <TextBlock Text="QUADRATIC TIME
COMPLEXITY • O(n²)"
                   FontSize="10"
                   Foreground="#9A854F"
                   HorizontalAlignment="Center"/>

        <Border Background="#FFF0B8"
                CornerRadius="12"
                Padding="12">
            <StackPanel Spacing="6">
                <TextBlock Text="Enter grid size (1–10)"
                           FontWeight="Bold"
                           Foreground="#493B1E"/>

                <TextBox x:Name="SizeInput"
                         Text="5"
```

```
                PlaceholderText="Enter a number"

TextChanged="SizeInput_TextChanged"
                Background="#FFFCF2"
                Foreground="#352D20"
                Padding="7"/>
        </StackPanel>
    </Border>

    <Border Background="#FFFFFF"
            BorderBrush="#EAC64D"
            BorderThickness="1"
            CornerRadius="12"
            Padding="12">
        <StackPanel Spacing="8">
            <TextBlock Text="YOUR STAR GRID"
                       FontSize="12"
                       FontWeight="Bold"
                       Foreground="#8A6B16"/>

            <TextBlock x:Name="PatternText"
                       FontFamily="Menlo"
                       FontSize="15"
                       Foreground="#493B1E"
                       TextWrapping="NoWrap"/>
        </StackPanel>
    </Border>

    <Border Background="#FFF0B8"
            CornerRadius="12"
            Padding="12">
        <StackPanel Spacing="5">
            <TextBlock Text="OPERATION COUNT"
                       FontSize="11"
                       FontWeight="Bold"
                       Foreground="#8A6B16"/>

            <TextBlock x:Name="StepsText"
                       FontSize="18"
                       FontWeight="Bold"
                       Foreground="#493B1E"/>
        </StackPanel>
    </Border>

    <Button Content="✦ GENERATE GRID ✦"
            Click="GenerateGrid_Click"
            Background="#EAC64D"
            Foreground="#352D20"
            FontWeight="Bold"
            Padding="10"/>

    <TextBlock Text="NESTED LOOPS • O(n²)"
               FontSize="11"
               Foreground="#9A854F"
               HorizontalAlignment="Center"/>

  </StackPanel>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;
using System.Text;

namespace Star12;

public partial class MainWindow : Window
{
    public MainWindow()
    {
        InitializeComponent();
        UpdateGrid();
    }

    public void SizeInput_TextChanged(object? sender,
TextChangedEventArgs e)
    {
        UpdateGrid();
    }

    public void GenerateGrid_Click(object? sender,
RoutedEventArgs e)
    {
        UpdateGrid();
    }

    private void UpdateGrid()
    {
        if (!int.TryParse(SizeInput.Text, out int size) ||
            size < 1 || size > 10)
        {
            PatternText.Text = "Enter a number from 1 to
10.";
            StepsText.Text = "Invalid input";
            return;
        }

        StringBuilder grid = new StringBuilder();
        int operations = 0;

        for (int row = 0; row < size; row++)
        {
            for (int col = 0; col < size; col++)
            {
                grid.Append("★ ");
                operations++;
            }

            grid.AppendLine();
        }

        PatternText.Text = grid.ToString();
        StepsText.Text = $"{operations} operations
({size} × {size})";
    }
}
```

Project 12: Bloom generator (O(2n)

1. Flower13.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
```

```
    <Nullable>enable</Nullable>

<ApplicationManifest>app.manifest</ApplicationManif
est>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
    <PackageReference
Include="AvaloniaUI.DiagnosticsSupport"
Version="2.2.3">
      <IncludeAssets Condition="'$(Configuration)' !=
'Debug'">None</IncludeAssets>
      <PrivateAssets Condition="'$(Configuration)' !=
'Debug'">All</PrivateAssets>
    </PackageReference>
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xa
ml"
        x:Class="Flower13.MainWindow"
        Title="Bloom Generator"
        Width="380"
        Height="560"
        MinWidth="350"
        MinHeight="500"
        Background="#FFF9F5"
        FontFamily="Arial">

    <ScrollViewer>
      <StackPanel Margin="20" Spacing="14">

        <TextBlock Text="a little garden"
                   FontSize="13"
                   Foreground="#9A7883"
                   HorizontalAlignment="Center"/>

        <TextBlock Text="Bloom Generator"
                   FontSize="25"
                   FontWeight="Bold"
                   Foreground="#805C66"
                   HorizontalAlignment="Center"
                   TextAlignment="Center"/>

        <TextBlock Text="✿  ❀  ✿"
                   FontSize="35"
                   Foreground="#D88FA2"
                   HorizontalAlignment="Center"/>

        <Border Background="#FBE7E9"
                CornerRadius="16"
                Padding="14">
          <StackPanel Spacing="8">
            <TextBlock Text="How many flowers?"
                       FontSize="14"
                       FontWeight="Bold"
                       Foreground="#805C66"/>

            <TextBox x:Name="FlowerInput"
                     Text="3"
                     PlaceholderText="Enter 1 to 6"

TextChanged="FlowerInput_TextChanged"
                     Background="#FFFCFA"
                     Foreground="#805C66"
                     Padding="9"/>
          </StackPanel>
        </Border>

        <Border Background="#FFFFFF"
                BorderBrush="#F0D5DC"
                BorderThickness="1"
                CornerRadius="16"
                Padding="14">
          <StackPanel Spacing="8">
            <TextBlock Text="YOUR BLOOM
PATTERNS"
                       FontSize="12"
                       FontWeight="Bold"
                       Foreground="#9A7883"/>

            <TextBlock x:Name="PatternText"
                       FontFamily="Menlo"
                       FontSize="12"
                       Foreground="#805C66"
                       TextWrapping="Wrap"/>
          </StackPanel>
        </Border>

        <Border Background="#E7EEDD"
                CornerRadius="16"
                Padding="14">
          <StackPanel Spacing="5">
            <TextBlock Text="POSSIBLE
PATTERNS"
                       FontSize="11"
                       FontWeight="Bold"
                       Foreground="#637653"/>

            <TextBlock x:Name="CountText"
                       FontSize="23"
                       FontWeight="Bold"
                       Foreground="#637653"/>

            <TextBlock Text="Every flower can bloom
or stay closed."
                       FontSize="12"
                       Foreground="#637653"
                       TextWrapping="Wrap"/>
          </StackPanel>
        </Border>
```

```
        <Button Content="✿  GENERATE BLOOMS ✿"
                Click="Generate_Click"
                Background="#E9B9C7"
                Foreground="#684753"
                FontWeight="Bold"
                Padding="11"/>

        <TextBlock Text="EXPONENTIAL TIME • O(2ⁿ)"
                   FontSize="11"
                   Foreground="#9A7883"
                   HorizontalAlignment="Center"/>

    </StackPanel>
  </ScrollViewer>
</Window>
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Interactivity;
using System.Text;

namespace Flower13;

public partial class MainWindow : Window
{
  public MainWindow()
  {
    InitializeComponent();
    UpdateBlooms();
  }

  public void FlowerInput_TextChanged(object? sender, TextChangedEventArgs e)
  {
    UpdateBlooms();
  }

  public void Generate_Click(object? sender, RoutedEventArgs e)
  {
    UpdateBlooms();
  }

  private void UpdateBlooms()
  {
    if (!int.TryParse(FlowerInput.Text, out int n) || n < 1 || n > 6)
    {
      PatternText.Text = "Please enter a number from 1 to 6.";
      CountText.Text = "Invalid input";
      return;
    }

    StringBuilder output = new StringBuilder();
    int totalPatterns = 1 << n;

    for (int pattern = 0; pattern < totalPatterns; pattern++)
    {
      for (int flower = 0; flower < n; flower++)
      {
        bool blooms = (pattern & (1 << flower)) != 0;
        output.Append(blooms ? "✿ " : "○ ");
      }

      output.AppendLine();
    }

    PatternText.Text = output.ToString();
    CountText.Text = $"{totalPatterns} patterns";
  }
}
```

Project 14: Petal Combinations

1. Flower14.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<ApplicationManifest>app.manifest</ApplicationManifest>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia" Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop" Version="12.1.4" />
    <PackageReference Include="Avalonia.Themes.Fluent" Version="12.1.4" />
    <PackageReference Include="Avalonia.Fonts.Inter" Version="12.1.4" />
    <PackageReference Include="AvaloniaUI.DiagnosticsSupport" Version="2.2.3">
      <IncludeAssets Condition="'$(Configuration)' != 'Debug'">None</IncludeAssets>
      <PrivateAssets Condition="'$(Configuration)' != 'Debug'">All</PrivateAssets>
    </PackageReference>
  </ItemGroup>


</Project>
```

2. MainWindow.axaml

```
using Avalonia.Controls;
using Avalonia.Controls;
using System.Text;

namespace Flower14;

public partial class MainWindow : Window
{
  public MainWindow()
  {
    InitializeComponent();
```

```
        UpdateCombinations();
    }

    private void PetalInput_TextChanged(
        object? sender,
        TextChangedEventArgs e)
    {
        UpdateCombinations();
    }

    private void UpdateCombinations()
    {
        if (!int.TryParse(PetalInput.Text, out int n) || n < 1
|| n > 6)
        {
            CombinationText.Text = "Enter a number from
1 to 6.";
            CountText.Text = "Invalid input";
            return;
        }

        StringBuilder result = new StringBuilder();

        for (int i = 0; i < (1 << n); i++)
        {
            for (int j = 0; j < n; j++)
            {
                result.Append((i & (1 << j)) == 0 ? "○ " : "✿
");
            }

            result.AppendLine();
        }

        CombinationText.Text = result.ToString();
        CountText.Text = $"{1 << n} combinations";
    }
}
```

3. MainWindow.axaml.cs

```
using Avalonia.Controls;
using Avalonia.Controls;
using System.Text;

namespace Flower14;

public partial class MainWindow : Window
{
    public MainWindow()
    {
        InitializeComponent();
        UpdateCombinations();
    }

    private void PetalInput_TextChanged(
        object? sender,
        TextChangedEventArgs e)
    {
        UpdateCombinations();
    }

    private void UpdateCombinations()
    {
        if (!int.TryParse(PetalInput.Text, out int n) || n < 1
|| n > 6)
        {
            CombinationText.Text = "Enter a number from
1 to 6.";
            CountText.Text = "Invalid input";
            return;
        }

        StringBuilder result = new StringBuilder();

        for (int i = 0; i < (1 << n); i++)
        {
            for (int j = 0; j < n; j++)
            {
                result.Append((i & (1 << j)) == 0 ? "○ " : "✿
");
            }

            result.AppendLine();
        }

        CombinationText.Text = result.ToString();
        CountText.Text = $"{1 << n} combinations";
    }
}
```

Project 15: LIttle Garden Explorer O(2n)

1. Flower15.csproj

```
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <OutputType>WinExe</OutputType>
    <TargetFramework>net10.0</TargetFramework>
    <Nullable>enable</Nullable>

<ApplicationManifest>app.manifest</ApplicationManif
est>
  </PropertyGroup>

  <ItemGroup>
    <PackageReference Include="Avalonia"
Version="12.1.4" />
    <PackageReference Include="Avalonia.Desktop"
Version="12.1.4" />
    <PackageReference
Include="Avalonia.Themes.Fluent" Version="12.1.4"
/>
    <PackageReference Include="Avalonia.Fonts.Inter"
Version="12.1.4" />
    <PackageReference
Include="AvaloniaUI.DiagnosticsSupport"
Version="2.2.3">
      <IncludeAssets Condition="'$(Configuration)' !=
'Debug'">None</IncludeAssets>
      <PrivateAssets Condition="'$(Configuration)' !=
'Debug'">All</PrivateAssets>
    </PackageReference>
  </ItemGroup>
</Project>
```

2. MainWindow.axaml

```
<Window xmlns="https://github.com/avaloniaui"
```

```
xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
        x:Class="Flower15.MainWindow"
        Title="Flower15 - Garden Explorer"
        Width="750" Height="600"
        MinWidth="600" MinHeight="500"
        Background="#FFF9F2">

    <ScrollViewer>
        <StackPanel Margin="30" Spacing="18">

            <TextBlock Text="🌷 GARDEN EXPLORER"
                       FontSize="13"
                       FontWeight="Bold"
                       Foreground="#71856A"/>

            <TextBlock Text="Let your garden grow"
                       FontSize="32"
                       FontWeight="Bold"
                       Foreground="#493D43"/>

            <TextBlock Text="Explore every possible flower arrangement."
                       FontSize="15"
                       Foreground="#82747A"/>

            <Border Background="#FFFFFF"
                    CornerRadius="20"
                    Padding="22"
                    Margin="0,8,0,0">
                <StackPanel Spacing="14">

                    <TextBlock Text="How many flowers?"
                               FontSize="18"
                               FontWeight="SemiBold"
                               Foreground="#493D43"/>

                    <TextBox Name="FlowerInput"
                             Text="3"
                             FontSize="18"
                             Padding="12"
                             Background="#FFF4F5"
                             Foreground="#493D43"/>

                    <Button Name="ExploreButton"
                            Content="Explore my garden 🌼"
                            Background="#D9A7B0"
                            Foreground="#493D43"
                            FontSize="16"
                            Padding="14"
                            CornerRadius="12"
                            Click="Explore_Click"/>

                    <TextBlock Name="ResultText"
                               Text="Your garden is waiting to bloom."
                               TextWrapping="Wrap"
                               FontSize="15"
                               Foreground="#71856A"/>

                </StackPanel>
            </Border>

            <Border Background="#F0F3E9"
                    CornerRadius="20"
                    Padding="22">
                <StackPanel Spacing="10">
                    <TextBlock Text="🌱 Garden notes"
                               FontSize="18"
                               FontWeight="SemiBold"
                               Foreground="#53664C"/>

                    <TextBlock Text="Each flower can bloom or rest."
                               Foreground="#53664C"
                               FontSize="15"/>

                    <TextBlock Text="The number of possible patterns doubles with every flower."
                               TextWrapping="Wrap"
                               Foreground="#53664C"
                               FontSize="15"/>

                    <TextBlock Text="Time complexity: O(2ⁿ)"
                               FontSize="17"
                               FontWeight="Bold"
                               Foreground="#53664C"/>
                </StackPanel>
            </Border>

        </StackPanel>
    </ScrollViewer>
</Window>
```

3. ***MainWindow.axaml.cs***

```
using Avalonia.Controls;
using System.Text;

namespace Flower15;

public partial class MainWindow : Window
{
    public MainWindow()
    {
        InitializeComponent();
        UpdateGarden();
    }

    private void PetalInput_TextChanged(
        object? sender,
        TextChangedEventArgs e)
    {
        UpdateGarden();
    }

    private void UpdateGarden()
    {
        if (!int.TryParse(PetalInput.Text, out int n) || n < 1 || n > 6)
        {
            FlowerDisplay.Text = "🌱";
            CountText.Text = "Check your number";
```

```
            ExplanationText.Text = "Enter a whole number
from 1 to 6.";
            GrowthText.Text = "";
            return;
        }

        int combinations = 1 << n;

        FlowerDisplay.Text = string.Concat(
            System.Linq.Enumerable.Repeat("✿ ", n));

        CountText.Text = $"{combinations}
combinations";

        ExplanationText.Text =
            $"{n} petals, each with 2 choices: bloom or
rest.\n" +
            $"2^{n} = {combinations} possibilities.";

        StringBuilder growth = new StringBuilder();

        for (int i = 1; i <= 6; i++)
        {
            growth.AppendLine($"{i} petals = {1 << i}");
        }

        GrowthText.Text = growth.ToString();
    }
}
```